Do brains synchronize more to biological motion? An inter-subject correlation (ISC) analysis of EEG whole-body movement data

People are highly sensitive to biological motion, the visual patterns associated with the movements of living entities. For example, we can recognize human movement from just a few moving dots placed on the major joints of the body, known as point-light displays (PLDs). At the neural level, the response to biological motion has been well-characterized within individuals, implicating regions of the visual and sensorimotor systems. However, less is known about the shared temporal characteristics of neural processing across individuals when viewing biological motion. Recent research using naturalistic movies suggests that dynamic visual stimuli can elicit shared patterns of neural activity across observers, or inter-subject correlation (ISC). Yet, whether PLDs of biological motion elicit shared neural processing across individuals remains unclear. This project will measure ISC in electroencephalography (EEG) data recorded during viewing PLDs of coherent, whole-body actions compared to "scrambled" versions of the same stimuli with the starting locations of the dots randomized. We will use MATLAB and code from dmochow's "Reliable Components Analysis" GitHub repository (https://github.com/dmochow/rca). Using this code, the data is cleaned and then the inter-subject correlation (ISC) is calculated. ISC is a number that shows how closely each participant's brain activity matches that of the other participants. Every participant gets one ISC value for each condition, and we compare the two with a paired t-test. Finally, a higher ISC for the coherent movement condition would mean people's brains synchronize more when they see human movement compared to the scrambled movement of the dots.

1. Introduction
    - Background: Explain biological motion: people recognize human movement from just a few moving dots (point-light displays). Explain inter-subject correlation (ISC) and how that will be used to measure how similarly different people's brains respond to the same stimulus, and why it is a useful way to study shared perception.
    - Research question: do people's brains respond more similarly when they watch a coherent point-light walker than when they watch scrambled dots?
    - Hypothesis: ISC will be higher for the coherent walker.

2. Ethical Considerations
    - Use of existing data - Prof. Harris agreed to share the data
    - Participant privacy: Data will be stored securely and participants have ID number
    - EEG is non-invasive and has minimal risk

3. Data & Methods
    - Data: EEG recordings and participants looked at two conditions (coherent vs. scrambled PLDs). Number of participants: 109
        - 108 trials:
            - 54 emotional: 18 different emotional movements,
                - 3 happy and 3 sad and 3 actors (18 total)

        - 54 neutral: 6 different neutral movements, marching, jumping, touching toes (2 actors)
- Cleaning of EEG data in MATLAB: loading the files and triggers, separating out each condition, sorting trials by stimulus, filtering
- ISC analysis: Explains correlated components analysis and how it gives one ISC value per participant and condition, plus scalp maps and ISC over time.
- Statistical analysis: paired t-test in R that compares each participant's ISC across the two conditions (and maybe Bayes factor?)

4. Results
    - Mean ISC for each condition; test results (p-value, Bayes factor, effect size); scalp maps showing where the shared response is strongest, and the ISC over time?

5. Discussion/Conclusion
    - Interpretation whether the results support the hypothesis
    - What does it say about how the brain processes human movement?
    - Limitations (separate spatial filters per condition, broadband EEG only, sample size) and future direction (look at mu-rhythm)

References/Appendix